Linking calculated and perceved accessibility to define equity standards

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Abstract

This is the abstract.

It consists of two paragraphs.

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1. Introduction

Transportation equity refers to the distribution of the burdens and benefits of the transportation system among the population (Karner et al., 2020, 2024; Pereira et al., 2017; Pereira & Karner, 2021). Burdens include health-related concerns arising from exposure to air pollution, noise, and traffic; sedentary lifestyles; and the risk of injury or death from collisions. In contrast, the primary benefit is the improvement of population accessibility (Pereira & Karner, 2021), understood as the ease of reaching (spatially distributed) opportunities and destinations (5–8), such as employment, healthcare, educational services, and other places for social interaction.

Evaluating accessibility to employment destinations is considered a key indicator for understanding the performance and social outcomes of a city (Bania et al., 2008; Bastiaanssen et al., 2021; Boisjoly et al., 2017; Jin & Paulsen, 2018; Johnson et al., 2017; Matas et al., 2010; Owen & Levinson, 2015; Shen, 1998), since employment provides the financial means to support other life activities (Merlin & Hu, 2017). Insufficient or inadequate accessibility can prevent individuals from participating in location-based activities they wish to engage in for reasons beyond their control, in a process described in the literature as Transport-Related Social Exclusion (TRSE) (Casas, 2007; Lucas, 2012; Paez et al., 2009; Preston & Rajé, 2007; Luz & Portugal, 2022). Insufficient accessibility, when combined with other forms of social and economic disadvantage, can result in transportation poverty (Casas, 2007; Litman, 2024; Lucas, 2012, 2019; Paez et al., 2009; Preston & Rajé, 2007): the failure of transportation and land use systems to provide access to essential services or employment (Luz & Portugal, 2022). Transportation poverty increases vulnerability among individuals who already face challenges associated with age (van Dülmen et al., 2022), income (Allen & Farber, 2019), race (Awaworyi Churchill, 2020), gender (Murphy et al., 2023), disability (Upham et al., 2022), among other factors (Kim et al., 2024).

Accessibility is one of the primary outcomes of spatial development (Páez et al., 2012). It is influenced by the spatial distribution of the population and opportunities and depends on the characteristics and performance of the transportation network, as a product of both transportation systems and land use patterns (Wee & Geurs, 2011). There is growing consensus among researchers and transportation planners that the goal of transport policy should be to improve accessibility. This contrasts with traditional planning

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approaches that have prioritized mobility patterns that stimulate private automobile use and higher speeds as solutions to congestion and long commuting times. Mobility-centered policies often focus on quantitative metrics, such as increasing the number of trips, raising average speeds, and reducing congestion, treating mobility as an end in itself. However, this approach is problematic because people rarely travel simply for the sake of moving; instead, they travel to access opportunities, activities, or other people at their destinations. Therefore, since the main goal is to enable people to access activities, land use and transport planning practices must be redesigned to prioritize accessibility.

With the gentrification of Canadian downtowns and the suburbanization of low-income populations, it is estimated that over one million Canadians live in transport poverty (Allen & Farber, 2019). However, the lack of standards for the equitable distribution of transportation benefits complicates efforts to address transportation poverty. Additionally, there is an absence of transportation data in the form of open data products to support urban planners (Arribas-Bel et al., 2021; Behrendt & Sheller, 2021; Páez, 2021), and transportation planners receive limited formal guidance in assessing the equity implications of their projects (Miller, 2018; Palm et al., 2023; te Brömmelstroet et al., 2016). Lastly, few studies explicitly link accessibility levels with employment outcomes, such as employment opportunities declined or accepted due to transport conditions (Soukhov, Anastasia et al., 2024), or individuals' perceptions - for instance, their opinion on how easy is to reach employment opportunities and whether jobs are available within an ideal commuting time. Combined, this results in a gap in the development and implementation of accessibility standards to ensure equity in transportation planning and policy.

Recently, the Mobilizing Justice Partnership released the first national, large-scale survey designed to understand TRSE and transport poverty in Canada. The National Survey on Transport Poverty (NSTP) collected responses from more than 27,000 people across the country, providing an unprecedented look at the transportation challenges Canadians face, especially among equity-deserving groups whose experiences are often overlooked. The survey is openly available to researchers, decision-makers, and the public, and offers information on household dynamics, respondents' perceptions, and individual-level socioeconomic data.

This study is driven by the following research questions: How can calculated job accessibility be related to perceived employment outcomes? And could a given level of accessibility serve as a standard or goal for equitable transport planning? To answer these questions, this study has the general objective of identifying key thresholds in the relationship between calculated job accessibility and people's perceptions of employment outcomes in order to define equity standards for transportation planning.

Our research advances the current understanding of the literature on the equitable distribution of transportation benefits by proposing a method that identifies calculated accessibility thresholds based on their relationship with perceived accessibility. Identifying minimum job accessibility levels in practice would facilitate the implementation of normative criteria that can guarantee standards to prevent equity-deserving groups from being excluded from the labour market. Furthermore, accessibility thresholds can provide policymakers with an understanding of how much accessibility is needed to meet the fundamental needs of the population, increasing the potential for government adoption and enhancing the comprehensibility and relevance of standards for community members and those experiencing unequal mobility.

We also present, in a transparent manner, the moral guidelines that support the distribution of accessibility. In addition, to facilitate collaboration and further analysis and to ensure the transparency and replicability of our study, we developed this paper using literate programming, with the data analysis code accessible through our GitHub page (https://github.com/dias-bruno/NSTP), in alignment with best practices in spatial data science (Arribas-Bel et al., 2021).

2. Theoretical Background

2.1. Equity Standards

2.2. Calculated Accessibility

Aiming to recover the spatial interaction aspect of accessibility, firstly introduced by X X, and posterior left by Hansen and others researchers, Soukhov recently proposed a new definition of accessility as

Returning to the spatial interaction from x and, soukov et al recently revised and proposed a new definition of accessibility as the potential for spatial interaction with opportunities.

2.3. *Job accessibility and employment outcomes*

Studies connecting employment outcomes to job accessibility date back to the 1960s, when John Kain (1968) proposed the spatial mismatch hypothesis while analyzing Detroit, Chicago, USA. Kain (1968) hypothesized that the low employment rates among Black Americans in inner cities resulted from employment decentralization and the difficulty faced by the Black population in moving out of these areas, primarily due to racial discrimination and segregation. Later, some other researchers observed that similar forces could affect poor white populations segregated in inner cities due to income rather than race (Bania et al., 2008).

Studies linking employment outcomes and accessibility have produced varying results in terms of statistical significance and the impact of accessibility variables on the dependent variable. Bania et al. (2008) examined the relationship between job access and labour market outcomes (employment, earnings, hourly wage, and weekly hours worked) for women leaving welfare in Cuyahoga County, Ohio, USA. The study found no significant effect for job access variables. Instead, education levels and prior work history were consistently more important in explaining outcomes.

Delmelle et al., (2021) also found no significant relationship between accessibility and employment outcomes in the Metropolitan Area of Charlotte, North Carolina, USA. The authors developed three models for different population subsets (all households, low-income households, and high-income households) and found that changes in accessibility had no significant impact on employment rates. They suggested that the spatial separation between lower-wage workers and their workplaces might already be relatively small, which would explain the absence of significant relationship between accessibility on the analyzed employment outcome. However, this study only analyzed car-based job accessibility, justifying the motorized mode corresponds to more than 90% of the trips made.

In a Canadian context, Pis (2019) aimed to understand the extent to which job accessibility affects the probability of employment among immigrants in Montreal and Toronto when compared to the non-immigrant population. The author created different logistic regression models, starting with only a transit-accumulated job accessibility measure as an explanatory variable and, step by step, added new socioeconomic and demographic variables to the other models, increasing the complexity of the new models and isolating the effect of accessibility. The results show that job accessibility has a weak effect on the chances of employment in both cities, and increasing levels of job accumulated accessibility are associated with decreasing probabilities of employment. In Montreal, there is a downward slope of the marginal effects for all models, meaning that there is a generally inverse relationship between transit-based job accessibility and employment outcomes in the city, or there are unobserved factors that influence the probability of employment at different levels of accessibility. In the case of Toronto, even though accessibility is associated with positive employment probabilities in the more complex model, the marginal effect of accessibility depends on other spatial factors, and increased accessibility to employment is associated with lower employment probabilities for immigrants and non-immigrants. Furthermore, the two sets of subpopulations consistently exhibit almost parallel slopes in the models. The difference in employment probability between immigrants and non-immigrants persists at all levels of accessibility, suggesting that the cumulative measure of accessibility to employment does not explain the difference in employment probability between subpopulations.

Some studies have identified mixed effects of accessibility on employment outcomes. Hu (2017) separated the population of Los Angeles metropolitan area into six groups according to income level to analyze the effect of accessibility. Job accessibility had no statistically significant effect on the employment status of the lowest income (household income below $ 25,000) or middle to high income (income above $ 75,000) groups. The results corroborate the hypotheses that low-income groups cannot take advantage of proximity to employment and that high-income groups can overcome spatial barriers regardless of accessibility to employment. For the lowest income group (below $ 25,000), having a car significantly increased the chances of employment, but job accessibility had no effect. However, job accessibility positively affected the employment situation of groups with household incomes between $ 25,000 and $ 75,000 (middle to low income). In this case, for middle-to low-income job seekers, living in areas with greater accessibility increases the likelihood of being employed. A one standard deviation change in job accessibility was associated with a 10.5% change in the probability of employment for the group earning between $ 25,000 and $49,999, and an 11.2% change for people earning between $ 50,000 and $ 74,999. The author explains these differences in the effect of accessibility between income groups due to the spatial distribution of this subpopulation within the study area. However, it is important to mention that Hu (2017) included accessibility to public transport only for the lowest income, since they rely more on cars than the other groups, with the other income groups being analyzed in terms of accessibility to the car. Merlin and Hu (2017), also studying Los Angeles, compared four accessibility measures: two non-competitive (cumulative opportunity and gravity-based) and two competitive (Shen and doubly constrained gravity-based). They found that the statistical significance and coefficient size of accessibility variables increased with higher educational levels, suggesting that job accessibility has a stronger association with employment status for higher education groups. The reason why accessibility measures are less important for lower-educated groups may be that these groups are more limited by non-spatial factors that impede their employment opportunities, such as qualifications and social skills. On the other hand, for workers with a higher level of education, the time and monetary costs of spatial separation itself may play a relatively more important role, so accessibility measures may have a greater impact on their decision and ability to enter the labor market. In my opinion, good findings come from this study: they identified that competitive accessibility measures showed higher significance levels and better model fit than non-competitive measures. This result emphasizes the importance of using competitive measures when linking accessibility to employment outcomes. Competitive accessibility measures presented the strongest relationship with employment outcomes across all educational groups, while non-competitive measures in several cases do not have a statistically significant association with the employment of disadvantaged education groups.

Another mixed effect was obtained by Boisjoly et al. (Boisjoly et al., 2017), when the authors explored the relationship between job informality and transit accessibility in the São Paulo Metropolitan Region. According to their results, for workers earning below the minimum wage, higher transit accessibility was associated with a lower likelihood of informal employment. For those earning above the minimum wage, car ownership was more relevant than transit accessibility in determining informal employment status (transit accessibility is not significant). The discrepancy in the statistical significance of accessibility to jobs by public transport is probably due to the fact that low-income workers are more dependent on public transport to get around. On the other hand, workers earning above the minimum wage are more likely to have other transportation options, i.e. access to a car, which makes them less dependent on public transportation. As a result, accessibility by public transport to jobs is not a significant indicator of informality for higher-income workers.

In contrast, (Bastiaanssen et al., 2021; Duarte et al., 2023; Jin & Paulsen, 2018; Johnson et al., 2017; Matas et al., 2010) found significant effects of job accessibility on employment outcomes. Johnson et al. (2017) found that shorter public transport times were associated with higher employment levels in England. They found a statistically significant relationship, suggesting that, all else being equal, areas with shorter public transport times are associated with higher levels of employment. The highest values are found in London and dense urban areas, with the lowest value in rural areas.

Jin and Paulsen (2017) used a difference-in-differences model to compare changes in accessibility between 2010 and 2017 in the Chicago Metropolitan Area. They found that changes in job accessibility (Shen) indices were statistically significant in predicting changes in unemployment rates. Negative coefficients for overall, Black, and low-income accessibility indicated that increased job accessibility was associated with reduced unemployment rates.

Bastiaanssen et al (2021) showed that better public transport accessibility (Hansen type) improved employment probabilities, particularly in metropolitan areas and low-income neighborhoods. Matas et al. (2010) found that public transport accessibility positively affected employment probability in Barcelona and Madrid, with the effect decreasing with higher educational attainment. Duarte et al (2023) found that found that job accessibility had a positive effect on participation in labour market for women, a negative effect for informal employment for women, and a positive effect for women and men in the probability of being employed.

3. Moral priciples to define accessibility standars

4. Materials and Methods

The methodology can be synthesized in six mains steps (Figure 1): modelling job accessibility, selecting population groups with lowest perceived accessibility, predicting employment outcomes using calculated accessibility, setting equity standards for transportation, defining priority geographical areas for policy interventions, and applying equity standards and measuring transportation equity.

4.1. Data

4.2. Study area

4.3. Accessibility

4.4. Modeling

4.4.1. Spatial Filtering

4.4.2. Linear Models

4.4.3. Categorical Ordinal Models

4.5. Equity Standards

4.6. FGTs

4.7. Areas for priority intervention

5. Results and discussions

6. Limitations

7. Summary and conclusions